

STRATEGIC RETENTION STRATEGY: SECURING THE FUTURE OF COMPANY A

Subtitle: Data-Driven Insights for Reducing Customer Churn

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Course: GCI World 2026 – Final Assignment

Agenda

- Understanding the Business Challenge.
- Dataset Overview & Preparation.
- Exploratory Data Analysis (EDA).
- Machine Learning Approach & Model Evaluation.
- Strategic Business Recommendations.
- Impact & Conclusion.

The Business Challenge

Current Situation: Company A is facing a churn rate of 49.5%.

Problem: High customer turnover is eroding revenue and increasing operational costs.

Goal: Shift from reactive firefighting to a proactive, data-driven retention strategy.

Dataset Overview

Analyzed a 360-degree customer dataset (100k+ records).

Data Sources: Combined Customer Profiles (Client.csv) with Usage History (Record.csv).

Objective: Gain deep insights into behavioral patterns linked to customer loyalty.

Data Cleaning & Preprocessing

Raw data presented significant challenges with missing values.

Methodology: Applied median imputation to ensure dataset integrity.

This ensures that our predictive model is built on solid, reliable foundations.

Exploratory Data Analysis (EDA)

Initial analysis reveals a balanced churn distribution.

Key Finding: There is a clear divergence in usage patterns between loyal customers and those likely to churn, providing a strong signal for our AI model.

Predictive Modeling Approach

Selected Algorithm: Random Forest Classifier.

Why? It excels at capturing non-linear relationships and complex interactions in human behavior data, which simpler models often miss.

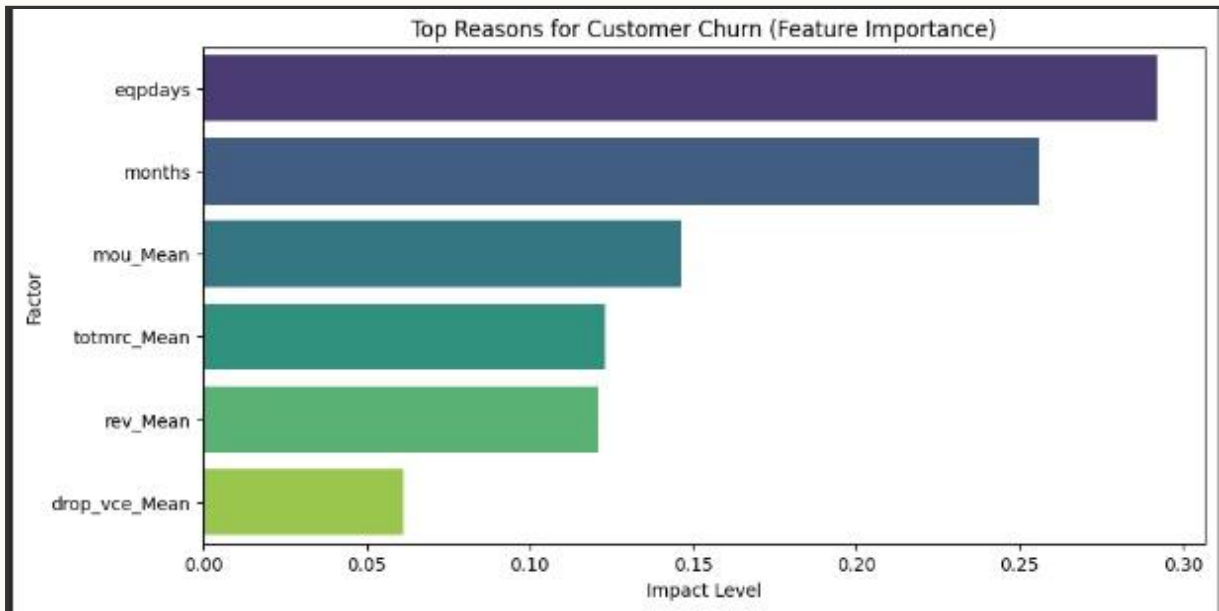
Model Evaluation

Primary Metric: ROC-AUC Score.

Achieved Score: 0.6487.

Interpretation: The model demonstrates a reliable ability to distinguish between "at-risk" customers and stable ones, validating our approach.

Key Drivers of Churn



Top predictors of churn:

- **Equipment Age:** Critical factor in customer satisfaction.
- **Tenure (Months):** Long-term usage trends.
- **Minutes of Use (MOU):** Declining usage acts as an early warning.

Deep Dive: Why Customers Leave

Insights: Long-tenured customers with devices older than 12 months are the most vulnerable segment.

Behavioral Signal: A downward trend in monthly minutes of use (MOU) is a strong indicator of a customer considering a competitor.

Recommendation 1: Loyalty Upgrade Program

Strategy: Target customers with outdated equipment (>12 months).

Action: Provide exclusive device upgrade discounts.

Expected Benefit: Revitalizes engagement and increases customer lifetime value.

Recommendation 2: Proactive Engagement

Strategy: Identify accounts with declining usage patterns.

Action: Offer personalized "Retention Bundles" (data or talk-time) to re-engage these segments.

Financial & Operational Impact

Efficiency: Shifting from broad marketing to targeted, data-backed campaigns will significantly lower acquisition costs.

ROI: High-accuracy targeting maximizes the return on marketing spend while preserving valuable revenue.

Conclusion

My analysis confirms that proactive churn reduction is achievable.

By leveraging these AI-driven insights, Company A can transform its customer experience and build sustainable, long-term loyalty.

References

Dataset: Company A provided GCI World 2026 materials.

Methodology: Final Assignment Tutorial & Course Resources.

AI Assistant: This analysis was supported by Gemini, an AI developed by Google.

Chat Link: [View Chat History](#)